

Introduction to Quantum Key Distribution
Advanced Certificate in Quantum Computing:
Algorithms and AI/ML



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Book:

- Ramona Wolf, *Quantum Key Distribution: An Introduction With Exercises*, Springer 2021

Introduction

- Cryptography:
 - Old concept (*fifth century BC*)
 - Codes and ciphers have evolved throughout history; there have always been attempts *to crack ciphers* and to come up with new and more secure ones.
 - Ciphers: *a secret way of writing*
 - In principle, they can be *cracked*.

Cipher text | P H H W P H D W W K H D S S O H E U H H

- Example of the encryption scheme:
 - *Caesar cipher*
 - The idea is *simple*: every letter of the alphabet is *shifted* by a certain number of letters.
 - Original form → every letter was shifted *three letters*

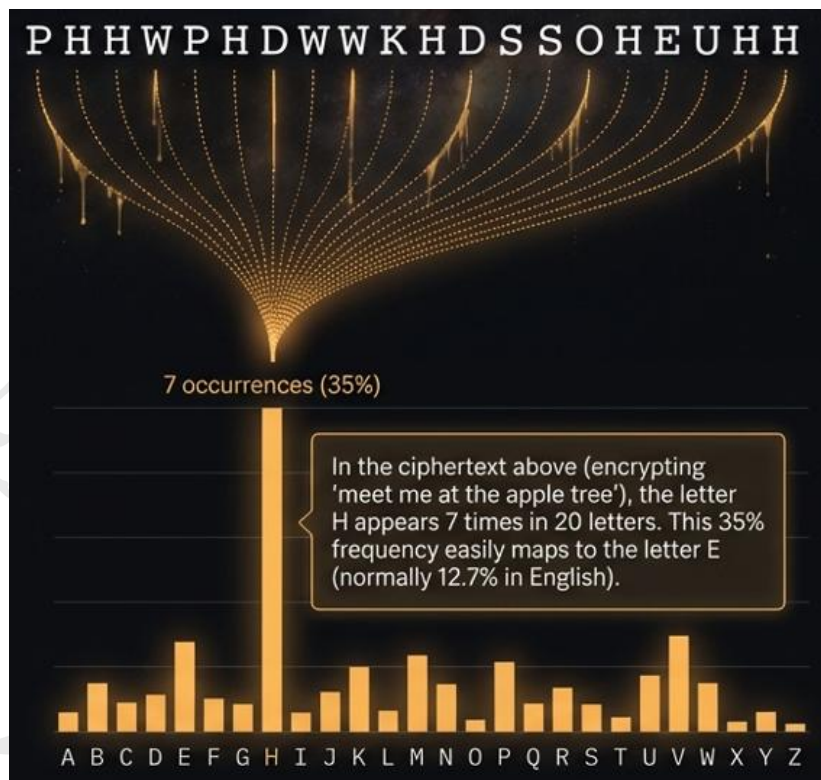
Plain alphabet	a b c d e f g h i j k l m n o p q r s t u v w x y z
Cipher alphabet	D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Caesar Cipher

Plain text		m	e	e	t	m	e	a	t	t	h	e	a	p	p	l	e	t	r	e	e
Cipher text		P	H	H	W	P	H	D	W	W	K	H	D	S	S	O	H	E	U	H	H

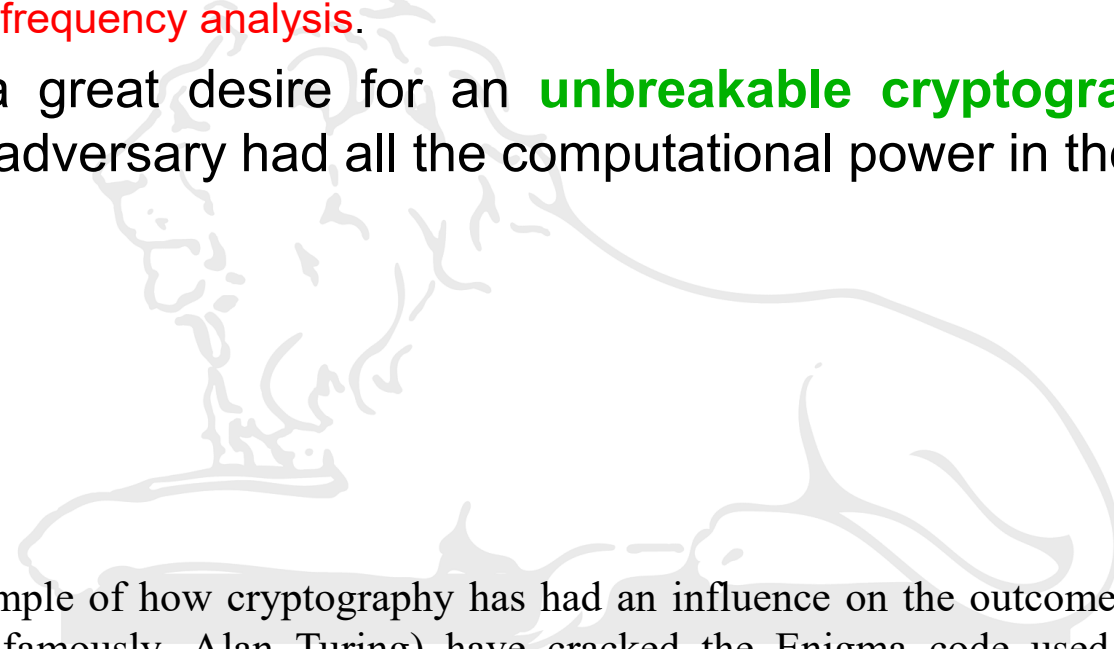
- Can we be **completely sure** that an adversary is unable to get the original message?
 - **Certainly not!**
 - For instance, if he knows about the Caesar cipher, he can **just reverse** the encryption process.
 - But even if he **does not know** about this encryption scheme, there is a way to recover the original message, which works by exploiting the structure of the English language
 - One can **analyze the frequency** with which certain letters appear in a typical English sentence, which is reflected in the frequency of certain letters in the ciphertext.
- In the English language, the letter “E” is the **most frequent one**.

Caesar Cipher

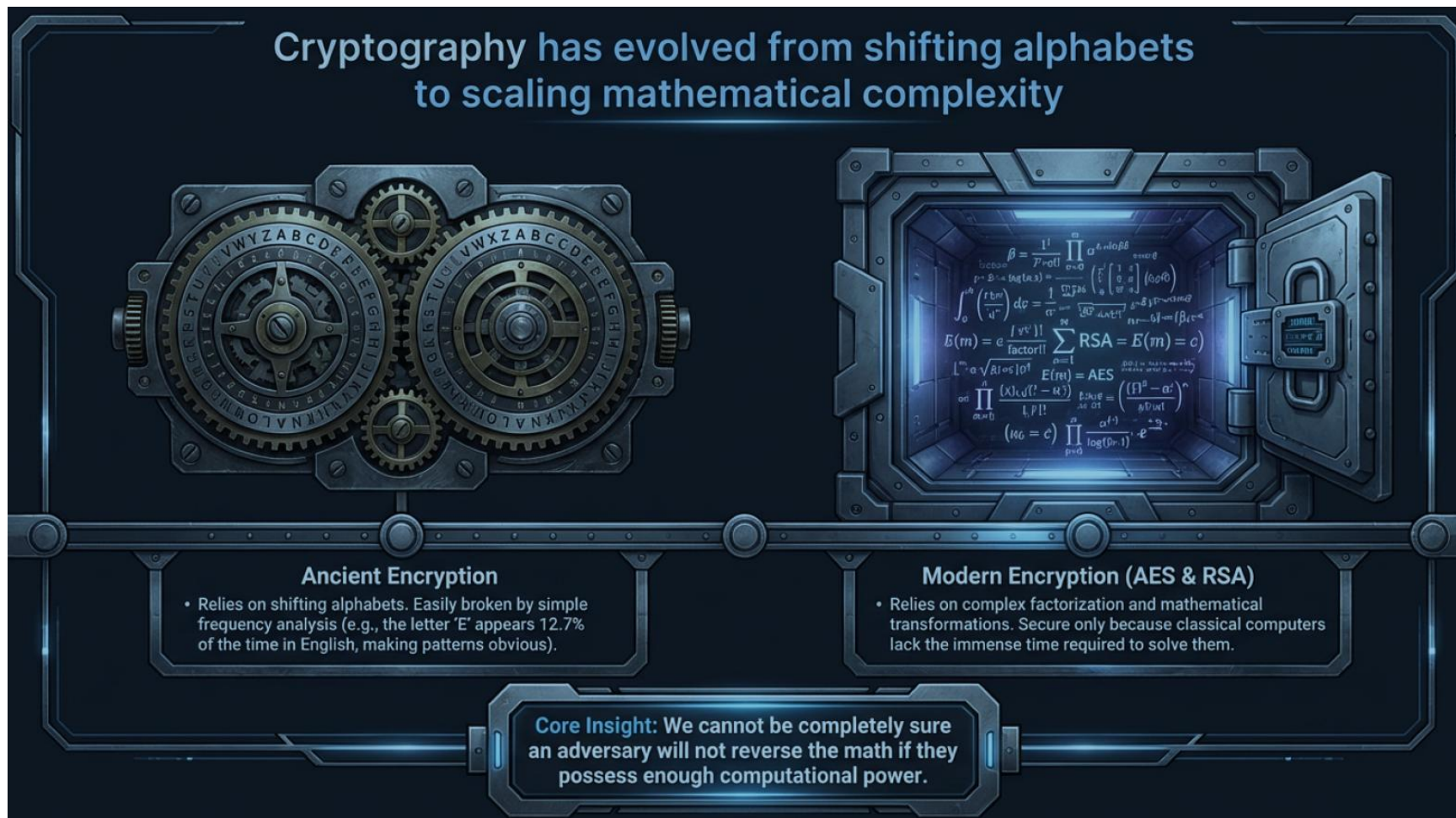


- The message above is a very good example of this:
 - The letter “H” appears seven times in a message of 20 letters, which corresponds to a frequency of 35%.
 - 12.7% for the letter “E” in the English language

Caesar Cipher

- There are **more complicated encryption schemes** than the Caesar cipher:
 - For instance, one could **shuffle the letters of the alphabet randomly** instead of just shifting them by a certain number.
 - However, this scheme can be cracked in the same way as the Caesar cipher, simply **by applying frequency analysis**.
 - There is a great desire for an **unbreakable cryptographic scheme**, even if an adversary had all the computational power in the universe.
- 
- A faint, light-colored background image of a classical statue, possibly a Greek or Roman figure, is visible behind the text.
- A famous example of how cryptography has had an influence on the outcome of a war is how the British (most famously, Alan Turing) have cracked the Enigma code used by the Germans to communicate during World War II.
 - Please watch: [Click Here](#)

Present Cipher Techniques



Current cryptography: Security based on **assumptions** about computational hardness

- Although **large-scale quantum computers** are not yet **available**, rapid progress suggests they may emerge within the **next decade or two**.
- Since current cryptographic systems rely on **computational hardness assumptions** that quantum algorithms can break, this creates **a long-term security risk**.

*we are in a “**pre-quantum threat**” phase*

Any solution?

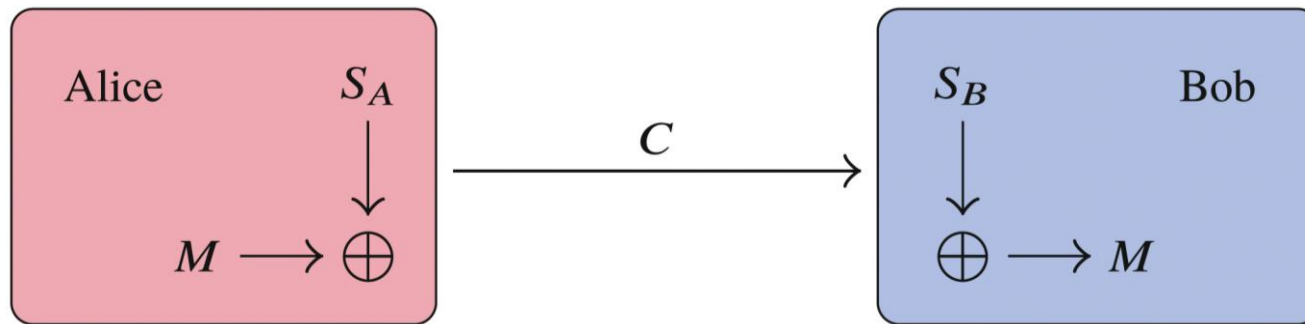
- In 1920s, an encryption scheme was proposed which is indeed provably secure:

Vernam cipher, also called one-time pad

One-time Pad Encryption

- Preliminaries:
 - Two parties, traditionally called **Alice and Bob**, want to exchange messages over a distance, and they want to be sure that *no one is able to read* these messages.
 - Hypothetical adversary → **Eve**
 - Alice and Bob need **a key to encrypt and decrypt messages** in a way that Eve cannot gain any information about the messages.
 - The word “**key**” now onwards refers to a **string of bits**, i.e., a sequence of 0s and 1s.
 - In the same fashion, whenever we say “**message**”, we mean a message which is in the form of a **bit string**.
- Alice and Bob have successfully created a pair of keys **S_A , S_B** , each of length n , and want to use it to send a message **$M = (M_1, M_2, \dots, M_n)$** of length n from Alice and Bob.
- $M_1, M_2, \dots, M_n$ represent the individual bits of the message given as a bit string.

Steps: One-time Pad Encryption



M : Message, S_A : Key at Alice, S_B : Key at Bob, and C : Ciphertext

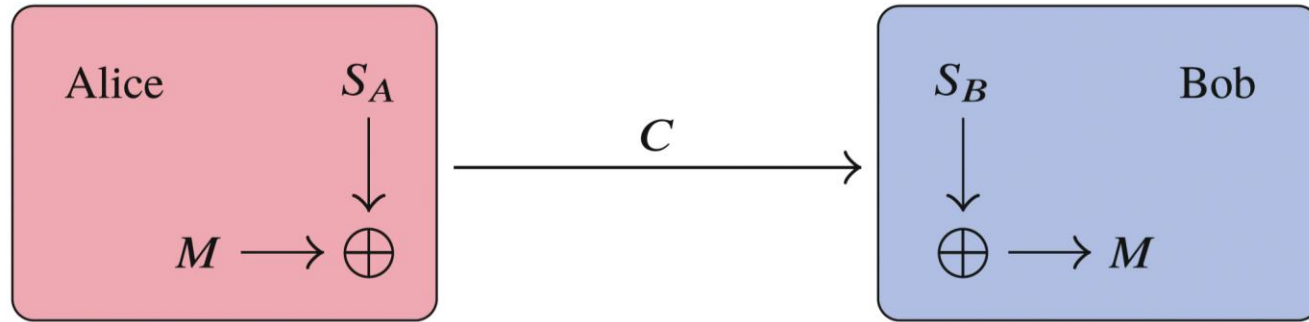
- Alice encrypts the message M with the key S_A by doing **binary addition**, which means that each bit of the ciphertext is produced by binary adding a bit from the key to a bit from the message:

$$C_i = M_i \oplus (S_A)_i$$

- The ciphertext $C = (C_1, C_2, \dots, C_n)$ is sent to Bob. This is done over a public channel, i.e., an adversary has full access to the ciphertext (but **cannot change the message**).
- Bob decodes the message by doing binary addition of the ciphertext and his key S_B :

$$M_i = C_i \oplus (S_B)_i$$

Steps: One-time Pad Encryption



- If the **keys are equal**, i.e., $S_A = S_B$, the message Bob holds after decoding is exactly the **same** as the one Alice sent.

$$\begin{aligned}
 M_i &= C_i \oplus (S_B)_i \\
 &= (M_i \oplus (S_A)_i) \oplus (S_B)_i \\
 &= M_i \oplus ((S_A)_i \oplus (S_B)_i) \\
 &= M_i \oplus ((S_A)_i \oplus (S_A)_i) \\
 &= M_i.
 \end{aligned}$$

Conditions to be met by the Key

- This encryption scheme is indeed information-theoretically secure, which means that the encrypted message (i.e., the ciphertext) **provides no information** about the original message (except the maximum possible **length of the message**), even though the adversary has complete access to the ciphertext.
- However, the key has to **meet certain properties**
 - It has to be **truly random**, which means that the individual key bits are not correlated in any way.
 - It has to be **at least as long as the original message**, which follows directly from the encryption process described above.
 - It can **never be reused in whole or partly**. Otherwise, an adversary can get information about the messages that are sent

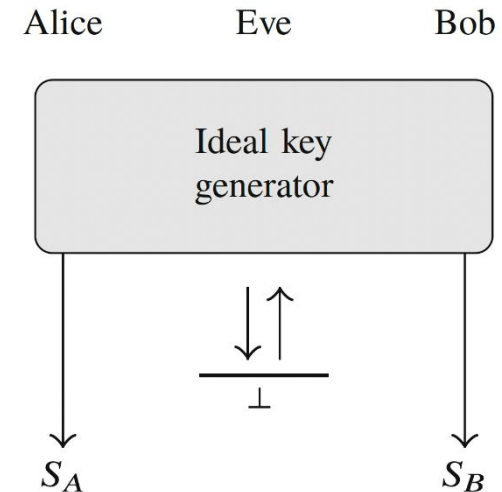
$$\begin{aligned}(C_1 \oplus C_2)_i &= M_i \oplus S_i \oplus N_i \oplus S_i \\ &= M_i \oplus N_i.\end{aligned}$$

- It has to be **kept completely secret**, i.e., the adversary has zero knowledge about the key.

Ideal Key Generator and Requirements

- Alice and Bob want a device that uses some sort of protocol to **generate keys** for Alice and Bob

- What are the **properties** that we require of such an ideal device?



- The key has to be **correct**, i.e., Alice and Bob should receive a **matching set of keys** to be able to communicate with each other:

$$S_A = S_B \equiv S$$

- The key should be **random**, i.e., every possible bit string of length l appears with the same probability.

$$\Pr[S = s] = 2^{-l}$$

- Eve **cannot interact** with the device

Practical Requirements

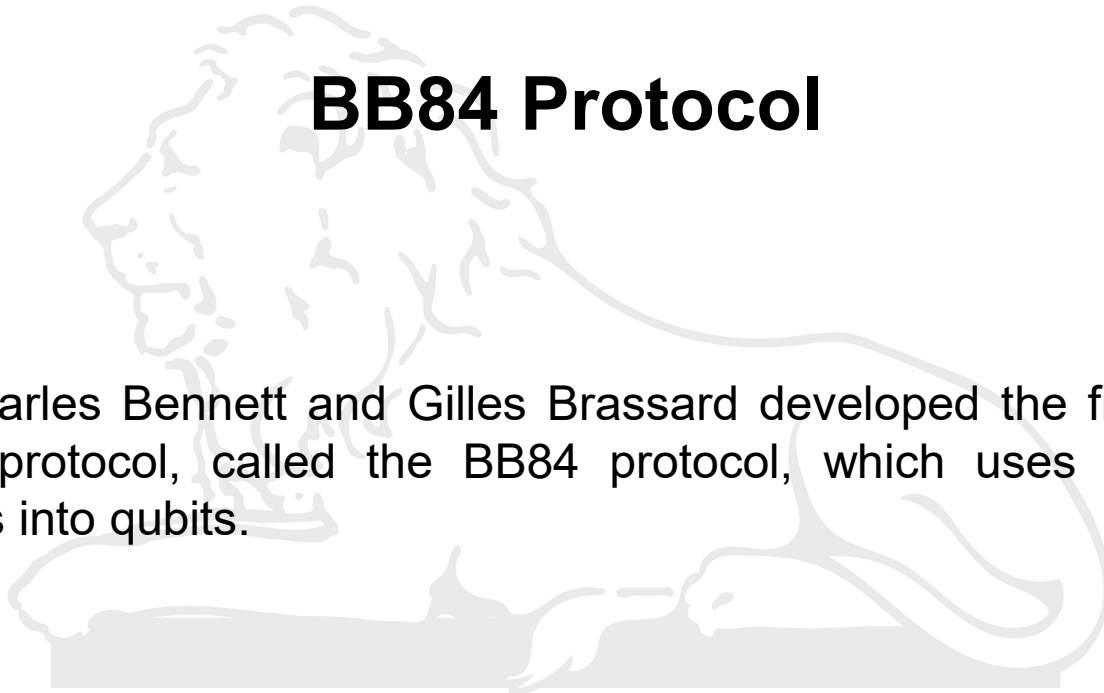
- Unfortunately, we will **never** be able to construct such an ideal key-generating device.
 - For instance, we **cannot completely prevent** Eve from interacting with the box.
- We can weaken the requirements so that the resulting device is not perfect anymore, but **its failure probability is small**.
- The requirements for such **a realistic device** are then changed in the following way:
 - We do not require that the keys the device outputs are always equal, but that the probability for the protocol not to abort and Alice's key differs from Bob's key is very small:
$$\Pr[S_A \neq S_B] \leq \epsilon$$
 - The generated key is close to a perfect key.
 - It is close to a uniformly random key
 - If Eve attack/interact the device, the device will abort the protocol and not output a key.
 - In this way, we lose the property that the device **always** outputs a secure key, but we can be sure that **if a key is produced, this key is secure**.

Quantum Key Distribution

- The **main goal** of quantum key distribution is to build **a device** where, whenever a key is generated, we can be sure that it is secure according to the previous **(weakened) requirements** so that Alice and Bob never use a corrupted key.

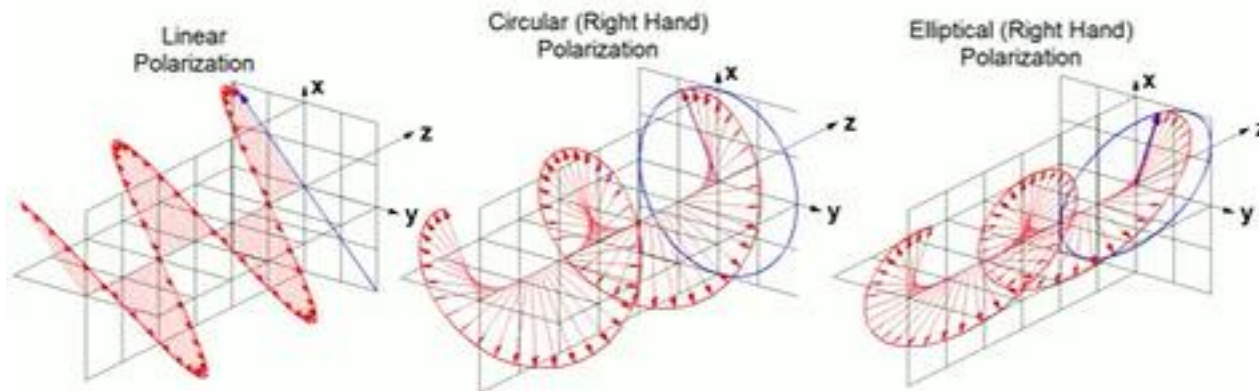
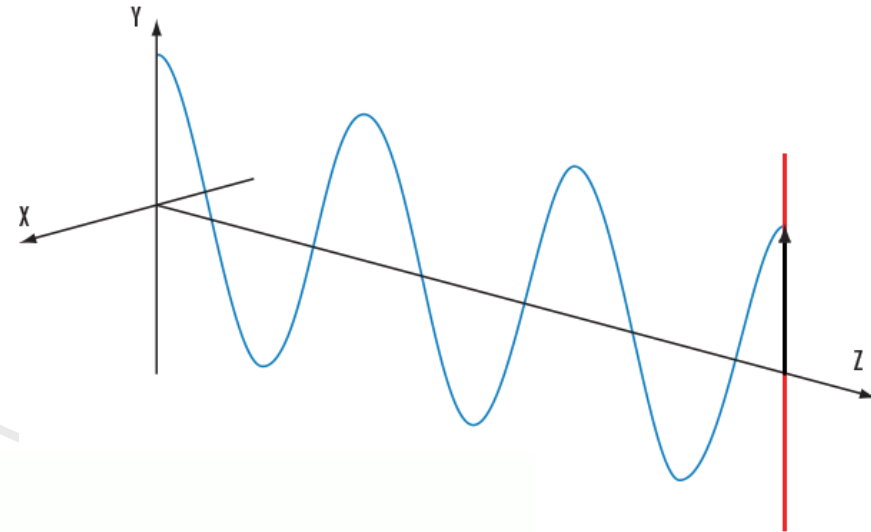
BB84 Protocol

- In 1984, Charles Bennett and Gilles Brassard developed the first quantum key distribution protocol, called the BB84 protocol, which uses the encoding of classical bits into qubits.



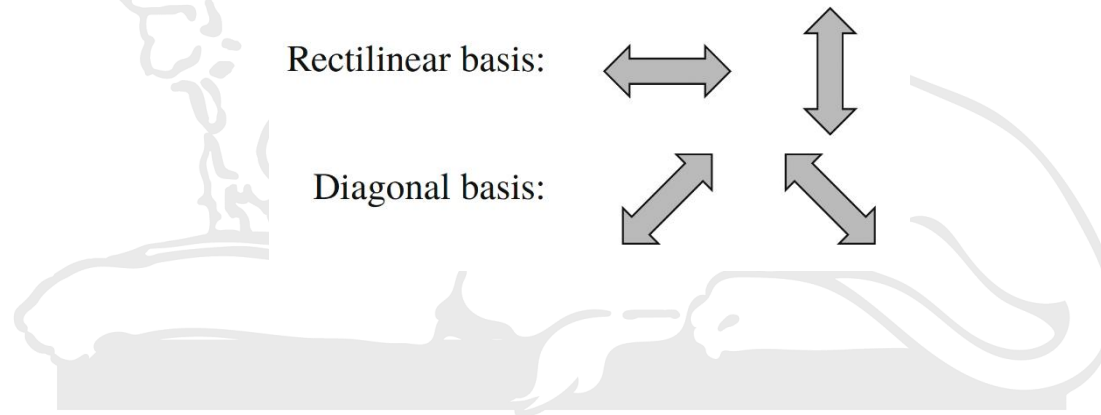
Polarization of Photons

The polarization of light specifies the geometrical orientation of the oscillation of the electromagnetic field associated with its wave.



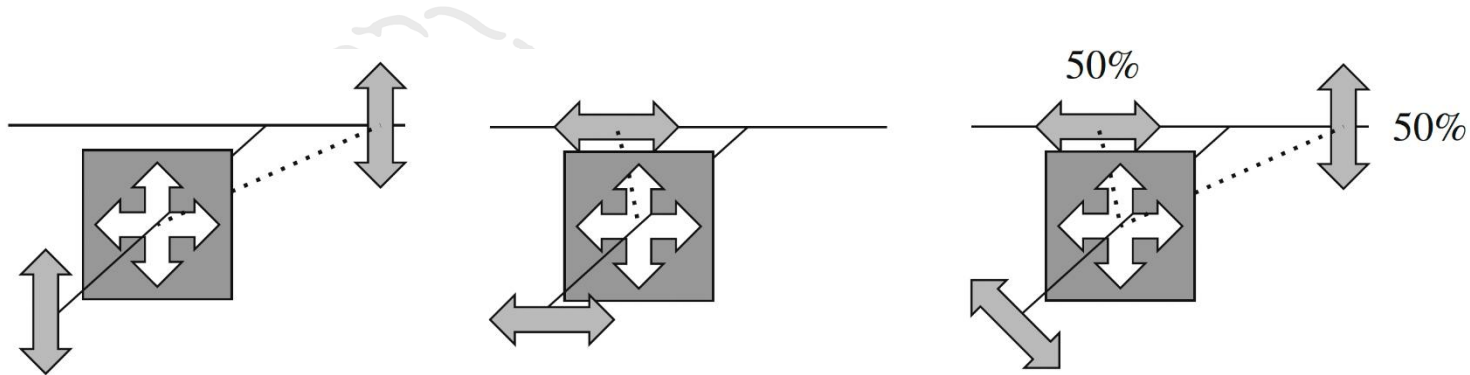
Polarizations for BB84

- We focus here on **linear polarization**, which means that the field only oscillates in one direction.
- We define two kinds of bases of linear polarization states:
 - The rectilinear basis, which includes **horizontal and vertical** orientations
 - The diagonal basis, which includes orientations **rotated by $+45^\circ$ and -45°** .



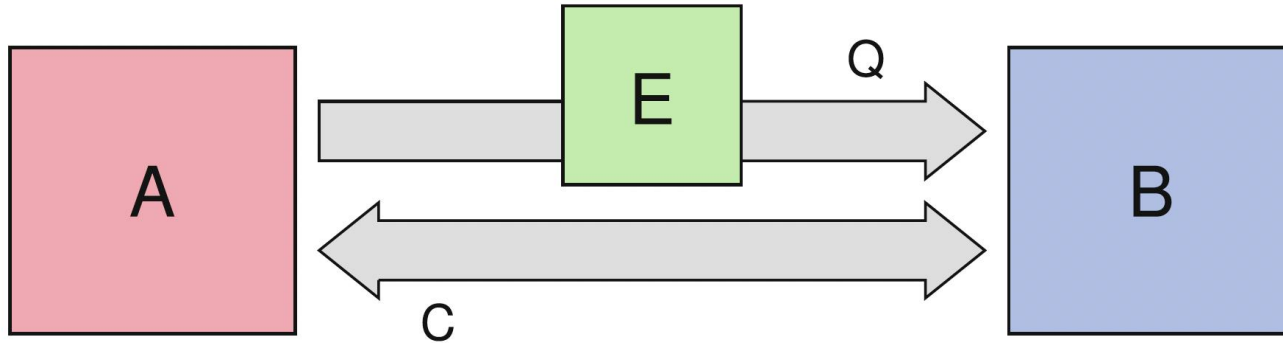
Polarization Filters

- There exist **filters** to distinguish between horizontally and vertically polarized photons (and, analogously, between $+45^\circ$ and -45° rotated photons), as depicted below:



- When a **vertically** (**horizontally**) polarized photon passes through a **rectilinear polarization filter**, it is deflected to the **right** (**left**), as depicted in the left and middle picture.
- When a photon that is polarized in the **diagonal basis** passes through a **rectilinear polarization filter**, it will be deflected to the left **50% of the time** and to the **right 50% of the time**, see the picture on the right

BB84 Protocol



Basic setting of the BB84 protocol.

- Two parties, Alice (A) and Bob (B), want to establish **a secret key** by using a **quantum channel** (Q) where an adversary, Eve (E), **can tap into without restrictions** and a classical channel (C) where Eve can only listen to the communication but cannot alter it.
- The protocol can be divided into two parts:
 - First, there is a **quantum transmission part**, where Alice and Bob prepare, send, and measure quantum states.
 - The second part is the **classical post-processing part** which is purely classical.

Quantum Transmission

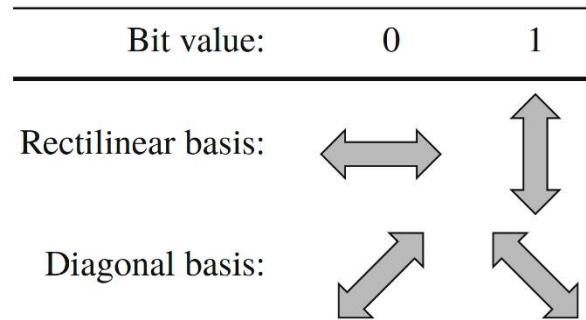
- It includes **encoding and decoding** of classical bits into quantum states and **communication** over a **quantum channel**.
 - Alice chooses a string of **N random classical** bits $X_1, \dots, X_N$.
 - Alice chooses a **random sequence of polarization bases**, where she can either choose the rectilinear basis (R) or the diagonal basis (D).
 - Alice **encodes** her bit string into a collection of photons with polarization according to the chosen bases, where the encoding works as shown below:

Bit value:	0	1
Rectilinear basis:		
Diagonal basis:		

- When Bob receives the photons he randomly (**and independently of Alice!**) decides for each photon whether to measure it in the rectilinear or the diagonal basis to obtain classical bits.
- After this step, Alice and Bob both hold a classical bit string, denoted $X = (X_1, \dots, X_N)$ for Alice and $Y = (Y_1, \dots, Y_N)$ for Bob.

This is called the **raw key pair**

Quantum Transmission

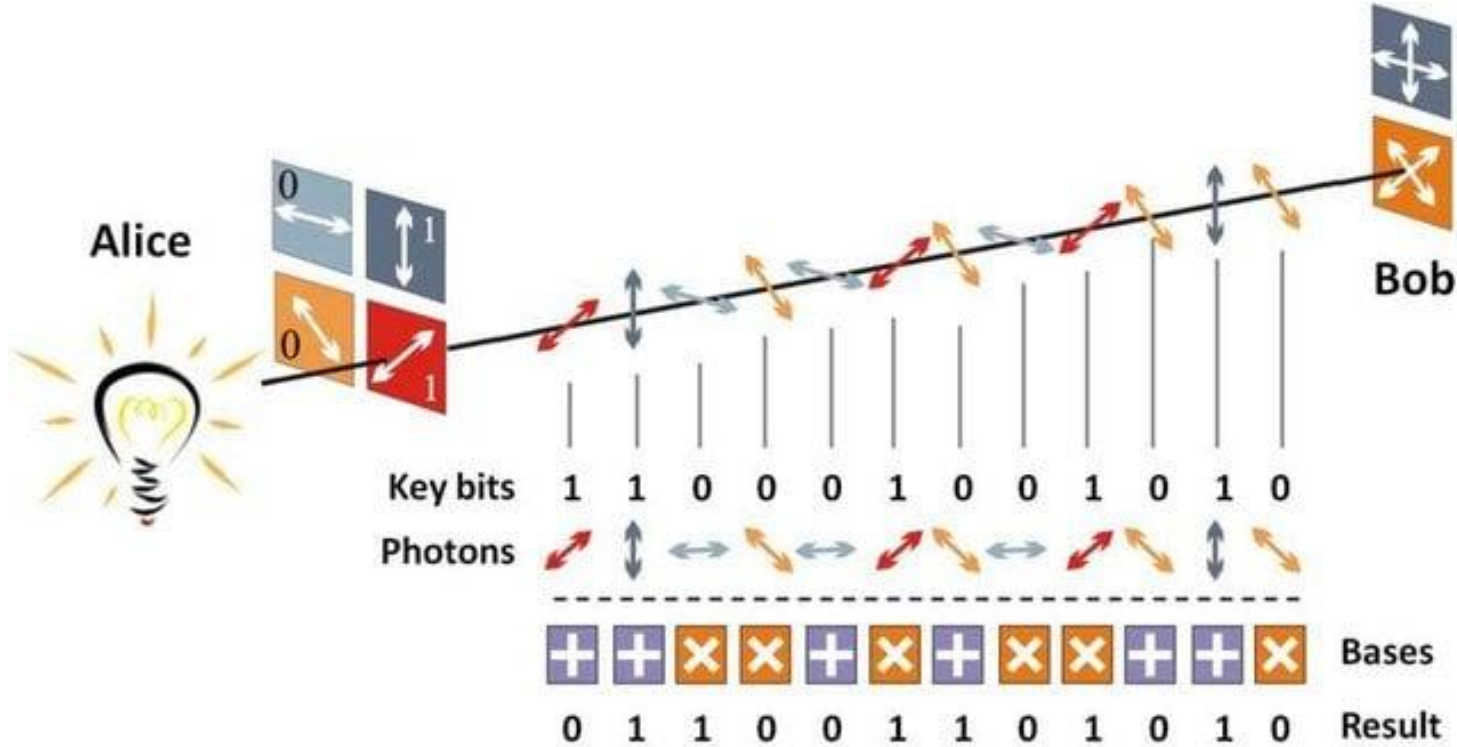


Quantum transmission

Alice's random bits	0	1	1	1	0	1	0	1	1	0	0	1	1
Random sending bases	D	R	D	D	R	D	R	R	D	D	D	R	D
Photons Alice sends													
Random receiving bases	D	R	R	D	R	D	D	D	R	D	R	R	D
Bits as received by Bob	0	1	0	1	0	1	1	1	0	0	0	1	1

Raw Keys

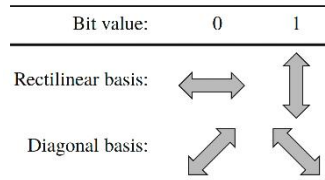
Quantum Transmission



Classical Post-Processing

- The rest of the protocol is **purely classical**.
- Alice and Bob exchange **a sequence of classical information** to **transform the bit strings they hold into a shared secret key**.
 - **Sifting Step:**
 - Bob publicly announces the bases he has chosen to measure the photons Alice has sent.
 - Alice compares Bob's bases to the ones she used and says which bases Bob has chosen correctly.
 - Alice and Bob discard all bits for which the encoding and measurement bases are not the same.
 - **Parameter Estimation (PE) Step:**
 - Alice and Bob want to compute a guess for the error rate in the quantum channel
 - Bob reveals some bits of his key at random. In case of no eavesdropping, these bits should be the same as Alice's bits and she confirms them.
 - If the error rate is too high, this indicates that there has been some eavesdropping and Alice and Bob abort the protocol.
 - The bits that have been revealed during this step are discarded afterwards as their information is now public to an eavesdropper.
 - **Information Reconciliation and Privacy Amplification (IRPA) Step:**
 - To further improve the secret key

Classical Post-Processing



Quantum transmission

Alice's random bits	0	1	1	1	0	1	0	1	1	0	0	1	1
Random sending bases	D	R	D	D	R	D	R	R	D	D	D	R	D
Photons Alice sends													
Random receiving bases	D	R	R	D	R	D	D	D	R	D	R	R	D
Bits as received by Bob	0	1	0	1	0	1	1	1	0	0	0	1	1

Classical post-processing

Bob reports bases of received bits	D	R	R	D	R	D	D	D	R	D	R	R	D
Alice says which bases are correct	OK	OK		OK	OK	OK				OK		OK	OK
Presumably shared information (without eavesdropper)	0	1		1	0	1				0		1	1
Bob reveals some key bits		1				1						1	
Alice confirms them		OK				OK						OK	

Sifted Keys

} PE

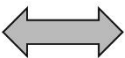
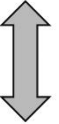
Outcome

Remaining shared secret bits	0			1	0					0			1
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IRPA on these keys

Security of the BB84 Protocol

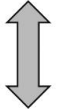
- How can Alice and Bob **recognize** that an **adversary is listening** to their communication while performing the steps of the BB84 protocol?

Bit value:	0	1
Rectilinear basis:		
Diagonal basis:		

Alice's bit:	0
Alice's basis:	D
State that is sent:	$\nearrow$
Eve's measurement basis:	R
State after the measurement:	$\leftrightarrow$
Bob's measurement basis:	D
Bob's bit:	1

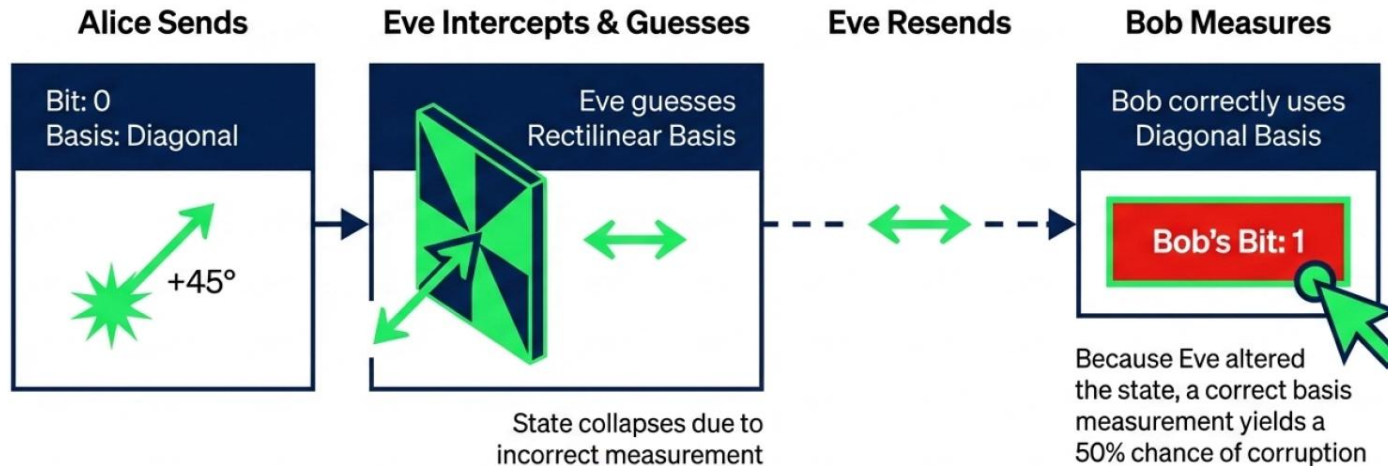
- In this case, Bob and Alice have used the **same bases**; hence, after the sifting step, they believe that they **are holding the same bit value**.
- Eve's interaction**, however, has **changed the quantum state** in a way that Bob's measurement has yielded **a different bit than the one Alice encoded**.
- In the **parameter estimation step**, where Bob and Alice compare parts of the actual bit strings they hold, **they will find that these bits do not match and reveal that an eavesdropper has tried to get some information**.

Security of the BB84 Protocol

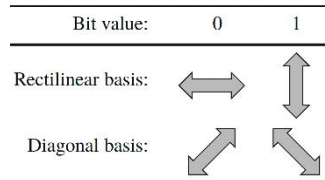
Bit value:	0	1
Rectilinear basis:		
Diagonal basis:		

Alice's bit: 0
 Alice's basis: D
 State that is sent: ↗
 Eve's measurement basis: R
 State after the measurement: ↔
 Bob's measurement basis: D
 Bob's bit: 1

The Anatomy of Eve's Error



BB84 Protocol with Eve



Key Bit	0	1	1	1	0	0	1	1	1	0
Alice Basis	R	R	D	R	R	D	D	R	D	D
Alice's state										
Eve's basis	D	R	R	D	R	D	R	D	R	D
Eve's state										
Eve's Result	0	1	1	0	0	0	1	1	1	0
Bob's basis	D	R	D	D	R	R	D	R	D	R
Bob's result	0	1	0	0	0	0	1	1	0	0
Sifting	X	✓	✓	X	✓	X	✓	✓	✓	X
Key bit		1	0		0		1	1	0	
PE		-			-				-	
Remaining Secret Key			0				1	1		

Quantum Performance Metrics

- Probability of Sift (P_{sift}):

$$P_{\text{sift}} = \frac{\text{\# of sifted bits}}{\text{\# of transmitted qubits}}$$

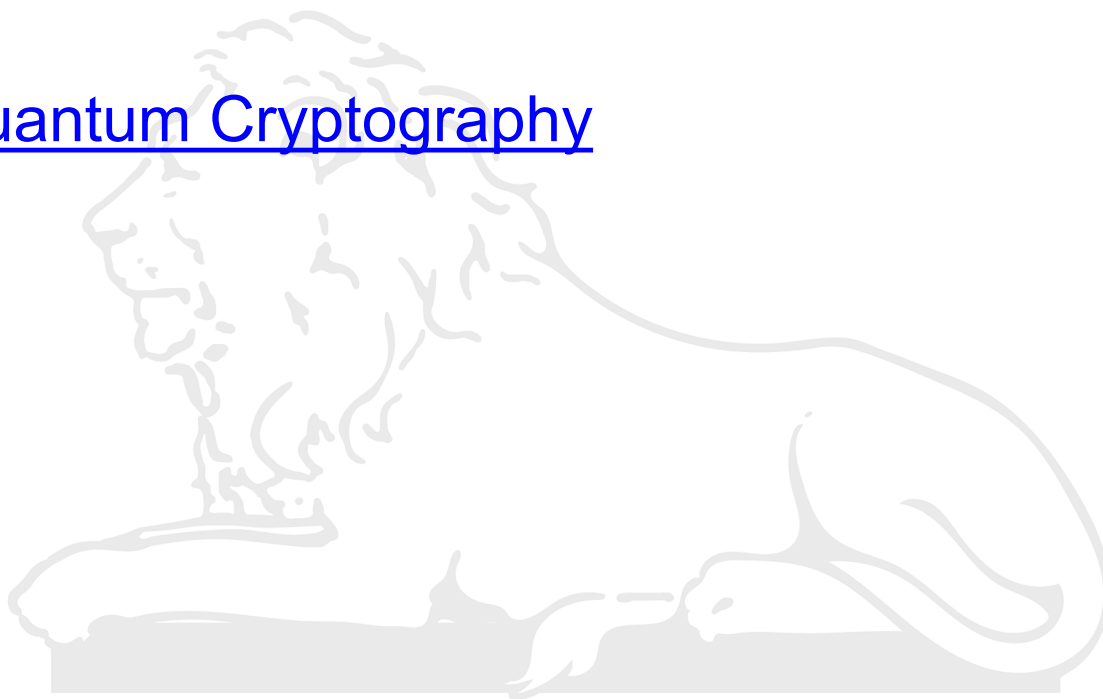
- Quantum Bit Error Rate (QBER):

$$\text{QBER} = \frac{\text{\# erroneous bits}}{\text{\# sifted bits}}$$

Revision through Simulation



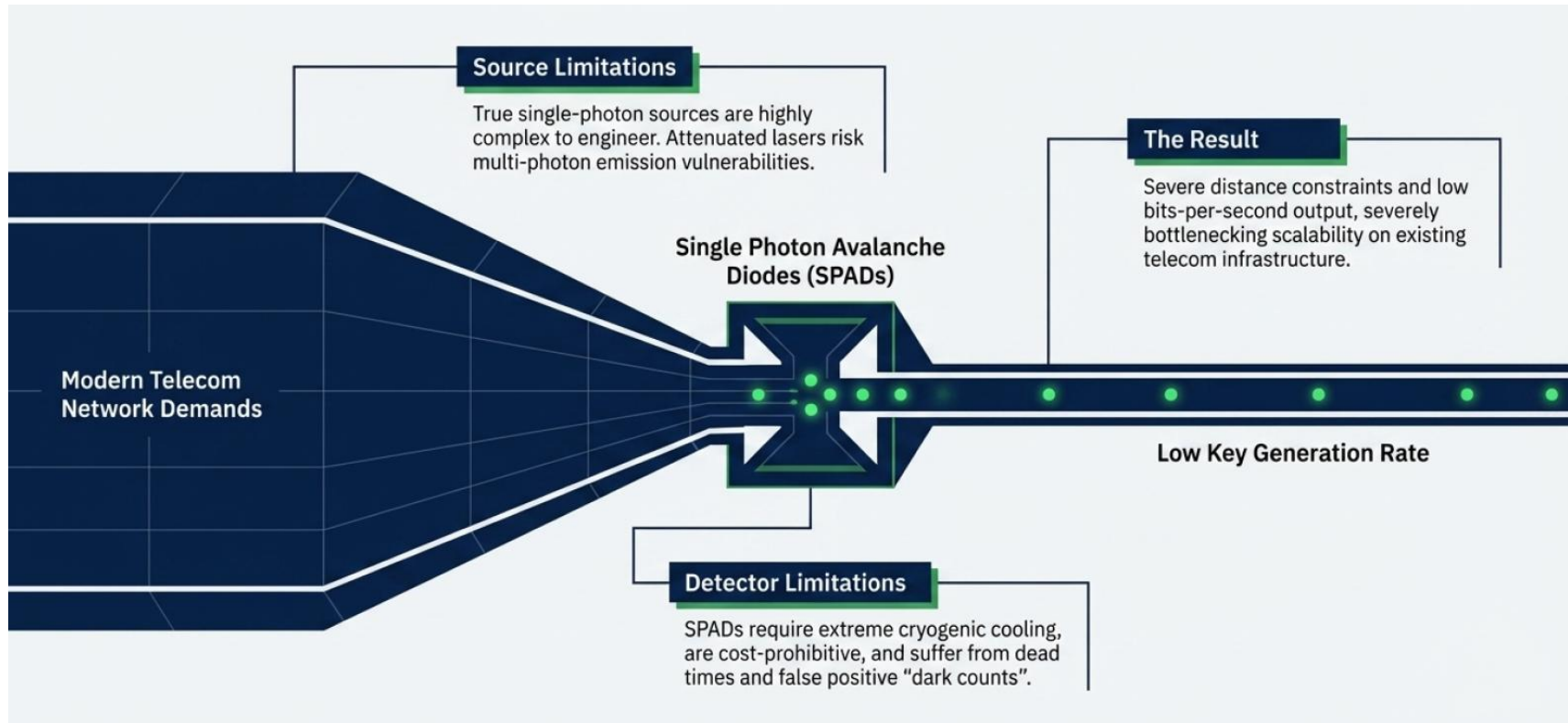
- [BB84: Quantum Cryptography](#)



Types of QKD

- Broadly divided into two types:
 - Discrete Variable QKD (DV-QKD)
 - Continuous Variable QKD (CV-QKD)
- BB84 comes under DV-QKD
- Some famous DV-QKD Protocols
 - E91 (Ekert protocol)
 - Proposed by Artur Ekert
 - B92: Proposed by Charles Bennett
 - Uses: Only two non-orthogonal states
 - 👉 Simpler than BB84
 - BBM92: Bennett, Brassard, and Mermin in 1992
- Discrete quantum states: Polarization, Phase, Time-bin, etc.

Limitations of DV-QKD



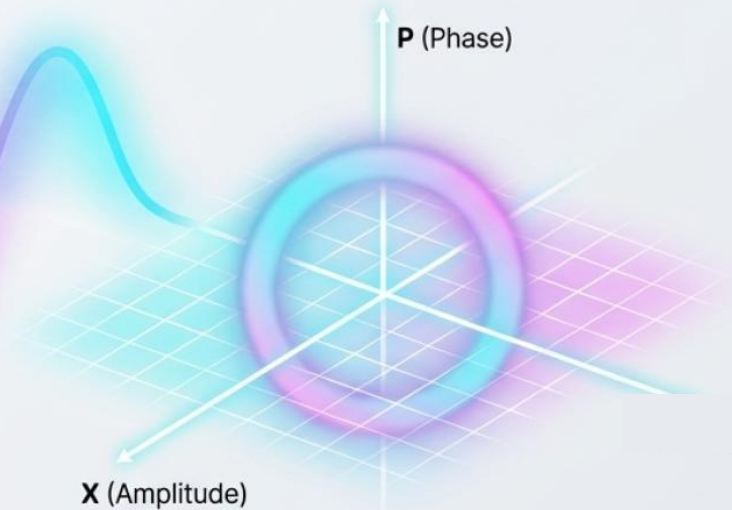
- Still, only the BB84-type is commercially implemented, i.e., modified forms (e.g., decoy-state BB84), not the *ideal textbook version*.
 - Links for information regarding commercial development (BB84-type):
 - <https://www.global.toshiba/ww/products-solutions/security-ict/qkd/products.html>
 - <https://www.quantum-info.com/English/product/quantum/2017/0901/324.html>

The Paradigm Shift

Instead of counting discrete, individual photons, CV-QKD encodes information into the continuous, macroscopic wave properties (quadratures) of a coherent laser field.

The Hardware Advantage

Bypasses the need for expensive single-photon detectors. CV-QKD natively integrates with standard, off-the-shelf telecom components like Homodyne and Heterodyne receivers operating at room temperature.



Difference between CV-QKD and DV-QKD

	 DV-QKD	 CV-QKD
Information Carrier	Single Photons	Coherent Laser States
Encoding Property	Polarization	Field Quadratures (Amplitude & Phase)
Detection Hardware	Single Photon Detectors (SPADs)	Homodyne / Heterodyne Receivers
Eavesdropping Metric	Quantum Bit Error Rate (QBER)	Excess Variance / Noise
Primary Advantage	Provable absolute physical security	High key rates & standard telecom compatibility

One of the first protocol: GG02 protocol (Gold standard)

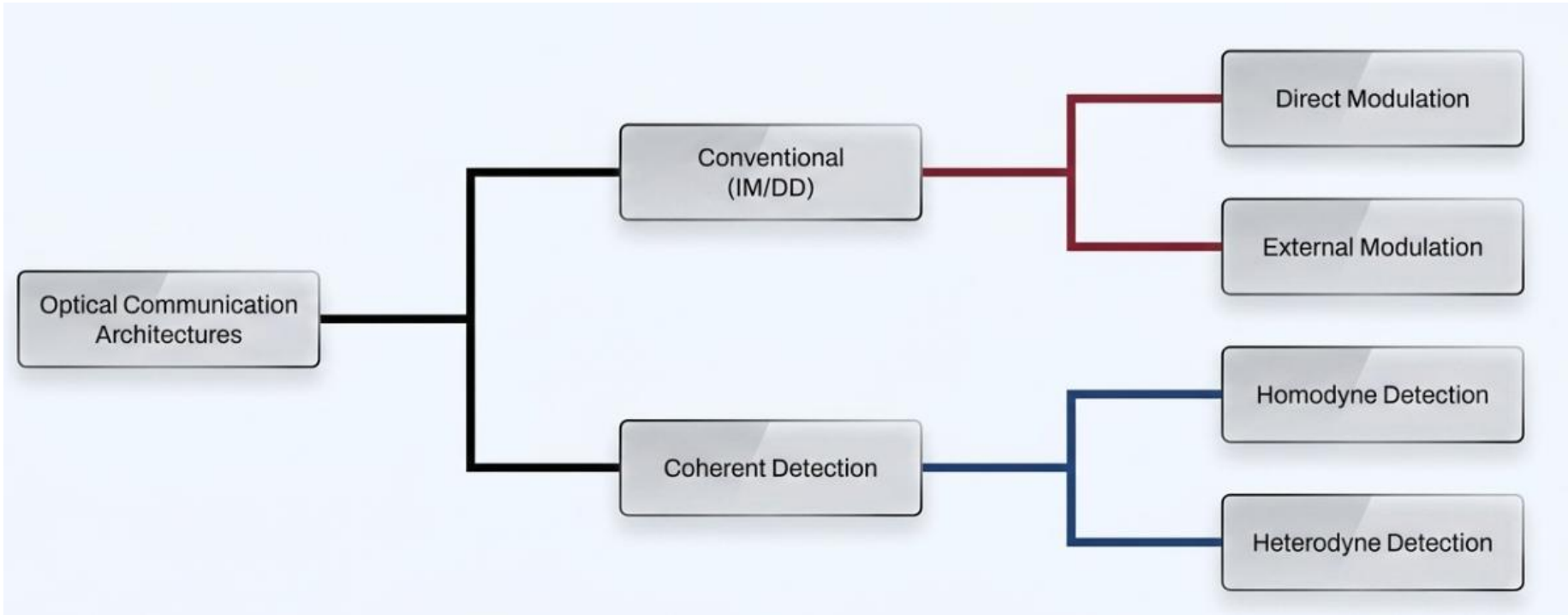
CV-QKD Timeline



Year	Milestone	Key Contribution
1999	First CV-QKD Proposal	Squeezed-state protocol introduced
2002	GMCS Protocol (GG02*)	Practical coherent-state CV-QKD
2003	Reverse Reconciliation	Enables long-distance operation
2007+	Security Proofs	Collective attack security
2010+	Composable Security	Finite-size analysis
2015+	DM-CV-QKD Emergence	Practical alternative to Gaussian modulation
2020+	High-speed & deployment	Real-time, field trials

*GGG02: **Grosshans-Grangier 2002**

Conventional Optical Communication Ecosystem



- IM/DD: Data center, access
- Coherent: Backbone, long-haul

IM/DD Optical Communication

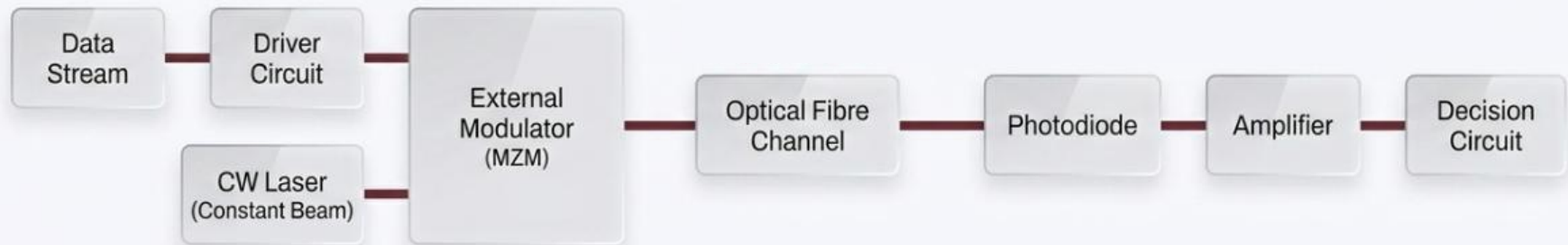
- Direct Modulation: 1–25 Gbps (typ.), up to ~50 Gbps

Direct Modulation



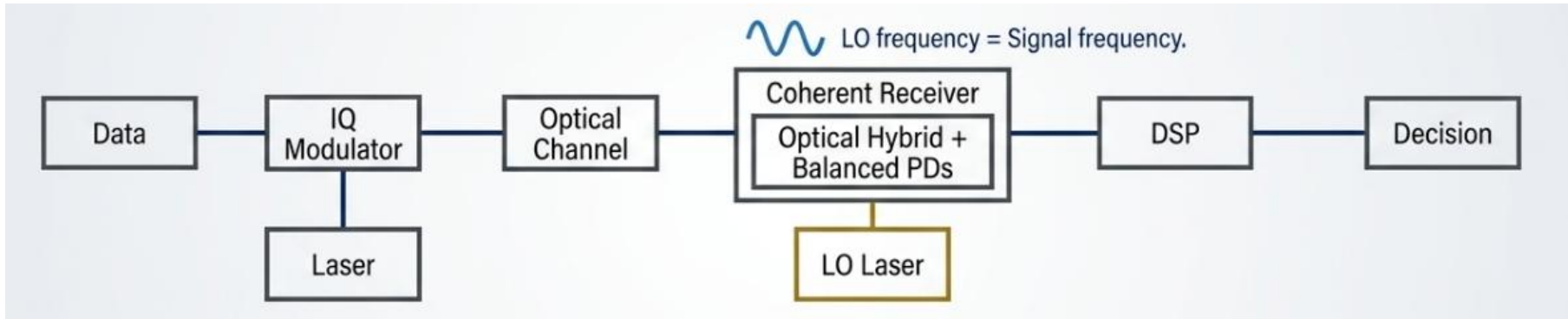
- External Modulation: 25–100+ Gbps

External Modulation

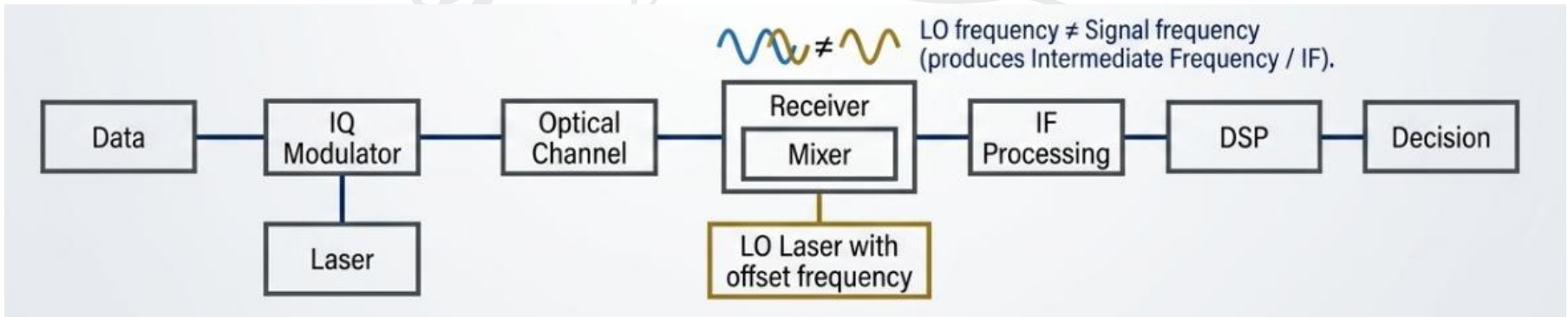


Coherent Optical Communication

- Homodyne Detection



- Hetrodyne Detection

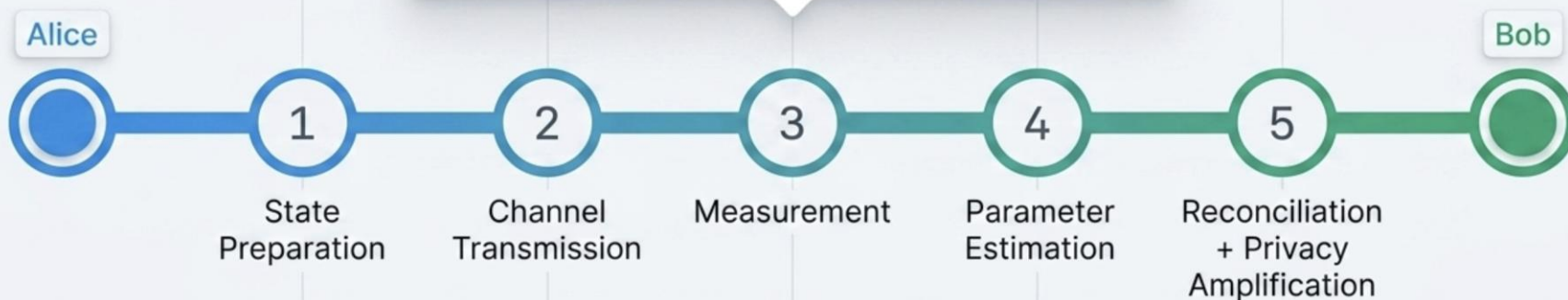


- Complexity: Homodyne > Hetrodyne
- Performance wise: Homodyne > Hetrodyne

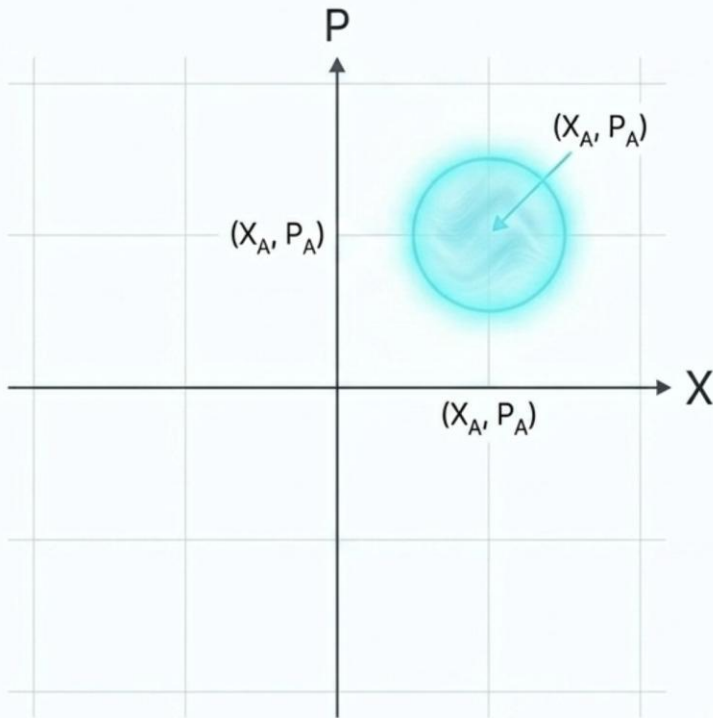
GG02 Protocol



The GG02 protocol serves as the gold-standard Gaussian implementation of this 5-step architecture, providing a rigorous mathematical framework for secure key extraction.



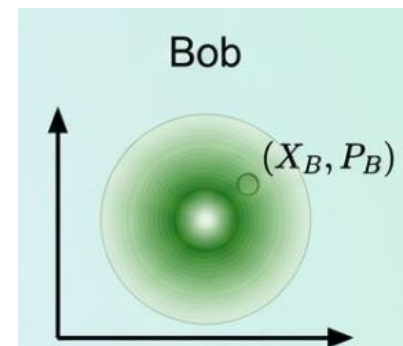
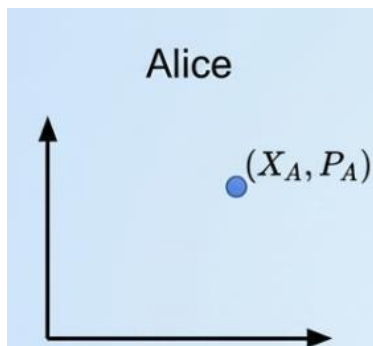
State Preparation



$$|\alpha\rangle = |X_A + iP_A\rangle$$

- Alice draws two random variables (X_A, P_A) from a Gaussian distribution with variance V_A .
- She prepares a coherent state.
- Information is encoded in continuous wave quadratures, bounded by the fuzzy circle of quantum uncertainty.

Channel Transmission



$$X_B = \sqrt{T}X_A + N_A \text{ and } P_B = \sqrt{T}P_A + N_B$$

$T = \text{Transmittance (Channel Loss)}$

Added Noise (σ_0^2) = Shot Noise + Excess Noise

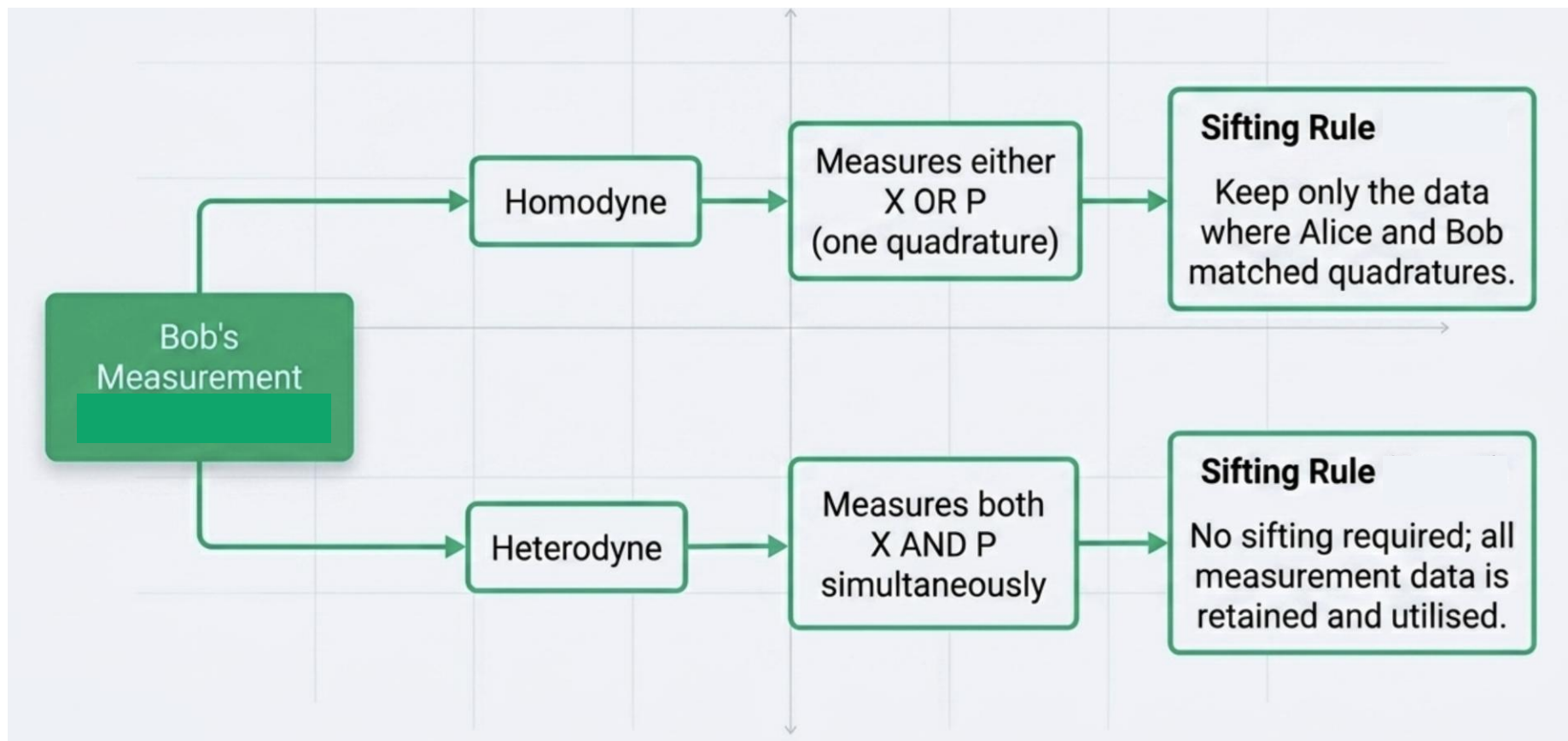
1. Shot Noise (Fundamental Noise)

- Originates from the discrete, quantum nature of light, where photon arrivals at a photodetector are random
- **Always present**

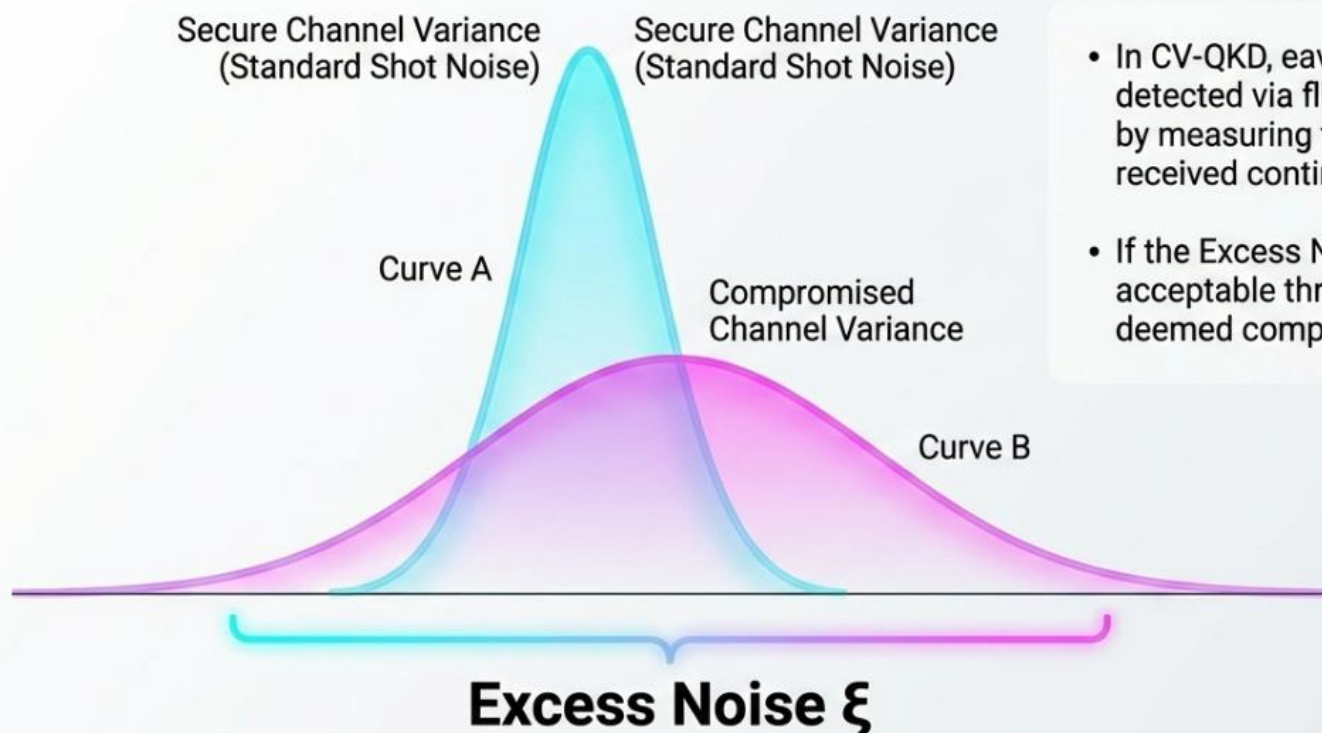
2. Excess Noise (ξ)

- Everything beyond shot noise
 - Laser intensity noise
 - Detector/electronic noise
 - Phase noise

Bob's Measurement



Excess Noise: The CV-QKD fingerprint of Eve.



- In CV-QKD, eavesdroppers are not detected via flipped bits (QBER), but by measuring the variance of the received continuous signal.
- If the Excess Noise (ξ) breaches the acceptable threshold, the channel is deemed compromised.

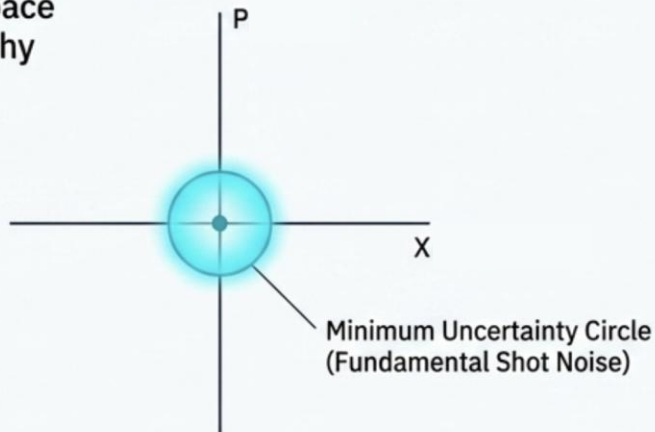
- Apart from excess noise, transmittance is also computed.

Security Guaranteed by Measurement Uncertainty



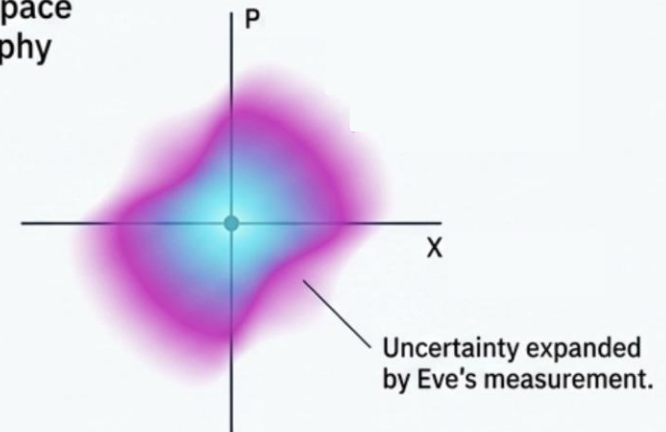
Secure Quantum State

Phase-Space
Topography



The Eavesdropper Effect

Phase-Space
Topography



Eve cannot simultaneously measure both amplitude and phase with perfect precision. Her interference physically smears the uncertainty circle, irrevocably adding quantum noise to the continuous signal.

Parameter Estimation

- Transmittance is derived from **signal correlation**, while excess noise reveals **any additional disturbance** in the channel.

$$X_B = \sqrt{T}X_A + N_A \text{ and } P_B = \sqrt{T}P_A + N_B$$

- Estimation of Transmittance:

$$T_X = \frac{\text{Cov}(X_A, X_B)}{(\text{Var}(X_A))^2} \text{ and } T_P = \frac{\text{Cov}(P_A, P_B)}{(\text{Var}(P_A))^2}$$

$$T_{est} = \frac{T_X + T_P}{2}$$

- Noise Estimation:

$$\text{Noise}_A = \text{Var}(X_B) - T_{est}\text{Var}(X_A)$$

$$\text{Noise}_B = \text{Var}(P_B) - T_{est}\text{Var}(P_B)$$

$$\text{Noise}_{est} = \frac{\text{Noise}_A + \text{Noise}_B}{2}$$

$$\xi_{est} = \text{Noise}_{est} - \sigma_0^2$$

Security Decision



Transmittance Consistency



$$|T_{est} - T_{expected}| < \Delta T$$

Does the measured channel loss match expected physical parameters?

Excess Noise Threshold



$$\xi_{est} < \xi_{th}$$

Is the anomalous disturbance below the critical security limit?

If both pass →
Proceed to Reconciliation +
Privacy Amplification

MATLAB Implementation

Limitations of Gaussian CV-QKD

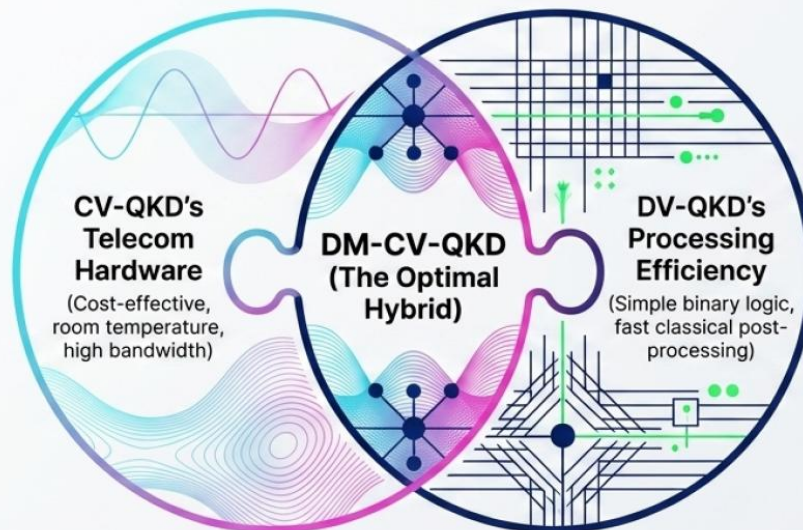
- Uses **continuous** Gaussian modulation of quadratures
 - Requires **high-resolution** analog processing
- Highly sensitive to:
 - **Calibration** errors
 - **System** imperfections
- Difficult integration with **digital communication systems**

Any solutions?

Discrete Modulation CV-QKD (DM-CV-QKD)



The Missing Link: Discrete Modulation CV-QKD



- Uses finite constellation (e.g., PSK, QAM)
 - Naturally compatible with digital processing
- Reduced system complexity
- Easier implementation and calibration